

Mohan Yang

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EDUCATION

University of California, Irvine

- Major: Data Science & Economics | Minor: Bioinformatics

Sep 2022 – Jun 2026

CGPA: 3.99/4.00

PUBLICATIONS

Yang, M.*, Tong, L. *, Guo, Z., Tan, Z., Holmes, T.C., Yu, Z., Xu, X. (2025). Evaluating the potential of acupuncture for Alzheimer's disease treatment: A meta-analysis and systematic review of mouse model studies. *Translational Psychiatry*.

Yang, M.*, Rodríguez Ortiz, C.*, Huynh, K.*, Grieco, S., Thakker, K., Yu, Z., Xu, X. (2025). MoSeq Reveals Behavioral Syllable Shifts after Ketamine in a CORT-Induced Depression Mouse Model. *In Preparation*.

Liu, J.*, **Yang, M.***, Nishida, K. A., Zhang, J. (2025). RNAFlux: A continuous normalizing flow model for RNA velocity. *In Preparation*.

Yang, M.*, Gaona Partida, P.R.*, Kanekar, A., Pantula, A., Chen, S., Kenney, A. (2025). Early Detection of Polycystic Ovary Syndrome Using the All of Us Research Program. *In Preparation*.

RESEARCH EXPERIENCE

UCI School of Medicine & School of Informatics and Computer Sciences

Irvine, CA

Advisors: Prof. Xiangmin Xu, Prof. Zhaoxia Yu

Apr 2024 – Present

Project 1: Meta-Analysis for evaluating the potential of electroacupuncture on Alzheimer's Disease

- Methods: Performed random-effects and network meta-analyses (R: *metafor*, *netmeta*); computed SMDs, ran subgroup/heterogeneity and publication-bias assessments, and visualized evidence with forest/funnel plots, network graphs, and Venn diagrams.
- Findings: Acupuncture improved cognition and reduced A β deposition, p-tau, and neuroinflammatory biomarkers in AD transgenic mice; Identified the most effective acupoints and found that multi-acupoint protocols do not uniformly outperform single-acupoint treatments.

Project 2: Web App Development - Automates extracting data from journal figures to support meta-analysis

- Built a full-stack ML-powered web application to automatically extract structured data (titles, bar heights, error bars, treatment groups) from journal plots, achieving 95%+ accuracy across 200+ test plots.
- Deployed in-lab to streamline data collection for meta-analysis, reducing extraction time from hours to minutes and standardizing dataset generation.

Project 3: MoSeq Analysis of Ketamine Response in a CORT Depression Mouse Model

- Processed 8 TB of depth video with MoSeq-PCA embeddings, AR-HMM tuning for syllable segmentation; built syllable transition matrices, clustered behavior modules, and ran Welch's t-tests with BH-FDR across cohorts/timepoints (fingerprint/transition heatmaps, network plots, dendrograms).
- Identified significant ketamine-induced behavioral changes, including shifts in rearing and grooming, and localized peak response windows and the transition circuits driving the effect.

UCI School of Informatics and Computer Sciences

Irvine, CA

Advisors: Junhao Liu, Prof. Jing Zhang

Dec 2024 – Present

Generative Model for RNA Velocity

- Challenged a common assumption in prior RNA-velocity models and relaxed it with a Neural ODE (continuous normalizing flow) decoder that captures coupled gene dynamics over continuous time.
- Built a variational generative model in PyTorch with a Zuko-based decoder to jointly infer latent cell time and model gene expression dynamics from scRNA-seq data.
- Reproducibility at scale: Tyro CLI + SLURM for structured experiments; Weights & Biases tracking; PyTest unit tests for data flows and training steps.
- Demonstrated that relaxing deterministic velocity assumptions enables more accurate reconstruction of gene-expression trajectories and smoother latent-time inference.

UCI School of Informatics and Computer Sciences

Advisor: Prof. Ana Maria Kenney

Early Detection of Polycystic Ovary Syndrome (PCOS)

- Leveraged the NIH All of Us Research Platform to extract and analyze 400K+ electronic health records (EHR) using SQL and pandas; designed and implemented large-scale data cleaning, preprocessing, and feature engineering on longitudinal clinical measurements and visit data.
- Proposed a Bayesian log-odds updating framework to estimate visit-level PCOS risk, sequentially accumulating longitudinal EHR evidence across patient encounters
- Incorporated polygenic risk score (PRS) as an individualized prior, enabling fusion of genetic risk with evolving clinical measurements.

Irvine, CA

May 2025 – Present

UCLA Neurology Department

Advisor: Prof. Jason Hinman

Axon Regeneration classification

- Built an R-based pipeline to process scRNA-seq data that performs quality control, filtering, normalization, variable feature selection, PCA/UMAP using Cell Ranger and Seurat.
- Applied Garnett classifier to categorize cells into regenerating, non-regenerating groups based on gene expression profiles. Quantified regenerative capacity across treatment groups and performed statistical hypothesis testing to assess group differences in regenerative outcomes.
- GitHub Link: https://github.com/HenryYang03/Axon_Regeneration

Los Angeles, CA

Sep 2024 – Apr 2025

UCI School of Informatics and Computer Sciences

Advisor: Prof. Weining Shen

GRU-Decay for Early Sepsis Prediction

- Explored the GRU-Decay model for imputing missing data in multivariate time series, focusing on early sepsis prediction using the PhysioNet Challenge 2019 dataset.
- Implemented the GRU-Decay mechanism using PyTorch, which dynamically adjusts feature imputation based on time gaps, ensuring missing values are weighted between prior observations and empirical means.
- Compared GRU-Decay against Simple GRU and LSTM models, demonstrating superior performance in AUC (0.9411) and accuracy (0.9426) for sepsis prediction.
- GitHub Link: https://henryyang03.github.io/SUROP_time_series/intro.html

Irvine, CA

June 2024 – Sep 2024

PROJECT

Ligand-Based Virtual Screening Model

- Developed GNN models to predict binding affinity for small molecules targeting the Amyloid protein, processed molecular data from the ChEMBL database, converting Canonical SMILES strings into molecular descriptors and graph representations using RDKit and PyTorch Geometric.

FELLOWSHIP & AWARDS

- Undergrad Research Fellowship (SURP), Multiple Time Series Integration with Different Resolutions (Summer 2024)
- Undergrad Research Fellowship (SURP), Automated extraction tool to Accelerate Meta-Analysis (Summer 2025)
- Undergrad Research Fellowship (SURP), Deep Generative Model for Cell Fate Dynamics Modeling (Summer 2025)
- Break Through Tech AI Fellowship at UCLA (2024–25 cohort, ~150 fellows nationwide)
- Computing Research Association (CRA) Outstanding Undergrad Research Award 2024-2025
- Donal Bren School of Information & Computer Science Honors Program

TECHNICAL SKILLS

Programming: Python, C/C++, Java, JavaScript, SQL, R, Stata

Tools: AI/ML (PyTorch, TensorFlow, Scikit-learn, Weights & Biases, Slurm, Pandas, NumPy), Web & Cloud (React, Flask, RESTful APIs, AWS, Docker), Dev Tools: (Git, PyTest, CLI configs)

LEADERSHIP EXPERIENCE

Learning Assistant for ICS-31 (Introduction to Programming)

- Led weekly office hours and review sessions; reinforced core Python concepts (data types, control flow, functions) and coached students on course projects. Collaborated with instructors to pinpoint common pain points and developed course-related practice activities and exam-prep materials.